Practice workbook · Protecting software by law: copyright, patents, trademarks and licences

Legal protections for software and systems (copyright, patents, trademarks, licences) and how licensing controls copying and distribution.

2 question parts: 1 worked-example retry and 1 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-16

1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1

2

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

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(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

(4)

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1. Question 6 · part d · 6 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

(d) Akiko wants to know how the software she writes can be protected.

Discuss how the software can be protected from illegal copying and/or illegal distribution.

You should consider:

- intellectual property
- licensing.

(6)

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(Total for Question 6 = 15 marks)

1. Question 6 · part d · 6 marks · Working and answer

Working and answer

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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5 Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

P AND (Q OR R)

You may not need to use all rows.

(3)

P	Q	R	Q OR R	P AND (Q OR R)

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

(2)

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

(2)

1 .....

2 .....

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2. Question 5 · part d · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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(d) The team needs to protect its intellectual property rights over the systems that it develops.

Give **three** types of legal protection the team could use.

(3)

1 .....

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2 .....

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3 .....

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(Total for Question 5 = 10 marks)

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Working and answer